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Sliding mode control of DC/DC converters

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Abstract

Sliding mode control algorithms for buck and boost power converters are surveyed in the paper. Current and voltage controls are demonstrated for the both cases. It is shown, that direct voltage control for a boost converter results in unstable zero dynamics. Chattering suppression based on harmonic cancellation principle along with switching frequency control is demonstrated.
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1. Introduction

ON–OFF is the only admissible operation mode for power converters, therefore sliding mode is an appropriate tool to control their currents and voltages, since this methodology implies discontinuous control actions. The main obstacle for sliding mode control implementation is a phenomenon called chattering in control literature [1–3] or ripple in power converter literature [4]. Due to high level of heat losses in power converters at high switching frequency and binary nature of control signals, the conventional methods of chattering suppression—use of asymptotic state observers [5] or adaptation with a time varying gain [6]—are not applicable.

The scope of the paper embraces, first, control algorithms for buck and boost converters [7] and, secondly, chattering suppression methods based on the harmonic cancellation principle.

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2. Buck DC/DC converter

The circuit structure of a *Buck* DC/DC converter is shown in Fig. 1, where the variables are defined as follows:

L: loop inductor; C: storage capacitor; R: load resistance; E: source voltage; I: input current; V: output voltage; u: switching signal taking value from the discrete set $U=\{0,1\}$.

The dynamic model *o*f the *Buck* type converter is given by

$$\dot{x}_1 = -\frac{1}{L}x_2 + u\frac{E}{L},$$

$$\dot{x}_2 = \frac{1}{C}x_1 - \frac{1}{RC}x_2, \tag{1}$$

with $x_1 = I$ and $x_2 = V$, control u can take two values 0 or 1.

The goal of control is to make an output voltage be equal to the desired value $V_d(t)$. The problem can be reformulated in terms of current control [7]: the state function u should be found such that current x_1 is equal to x_1^*.

If it happens, then mismatch $\Delta x_2 = V_d - x_2$, governed by equation $\Delta\dot{x}_2 = -(1/RC)\Delta x_2$, tends to zero.

The control is selected in the form

$$u = \frac{1}{2}(1 - sign(s))$$

to enforce sliding mode in

$$s = x_1 - x_1^* = 0$$

The sliding mode exists if the values s and

$$\dot{s} = \left[\frac{1}{L}\left(\frac{1}{2}E - x_2\right) - C\ddot{V}_d - \frac{1}{R}\dot{V}_d\right] - \frac{1}{2L}E\,\mathrm{sign}(s)$$

have opposite signs. For V_d being constant, the sliding mode can be enforced if $0 < V_d < E$. It means that the output voltage for a *Buck* converter is smaller than the source voltage.

Figs. 2 and 3 show the simulation results of proposed control algorithm for the Buck DC/DC converter. In the simulation, the converter parameters are selected as: $E = 20$ V, $C = 4\ \mu F$, $R = 40\Omega$ and $L = 40$ mH. The desired output voltage is $V_d = 7$ V. As can be seen from the figures, both the inductor current and the output capacitor voltage converge rapidly to their reference values.

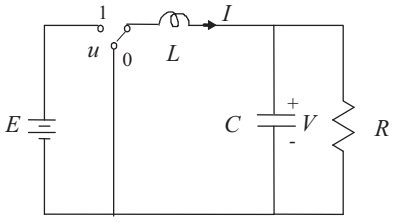


Fig. 1. Buck DC–DC converter.

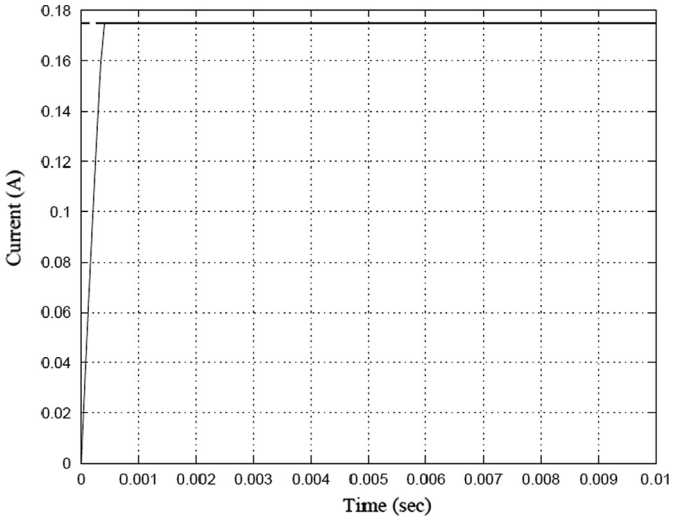


Fig. 2. Current response of a sliding mode controlled a buck DC/DC converter.

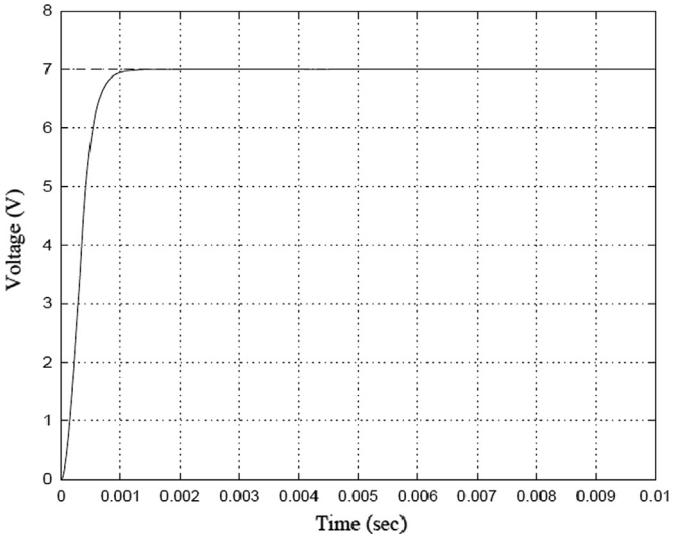


Fig. 3. Voltage response of a sliding mode controlled a buck DC/DC converter.

3. Boost DC/DC converter

Fig. 4 shows the principle of a *Boost* type converter, where the variables are defined as follows:

L: loop inductor; C: storage capacitor; R: load resistance; E: source voltage; I: input current; V: output voltage; u: switching signal taking value from the discrete set $U = \{0, 1\}$.

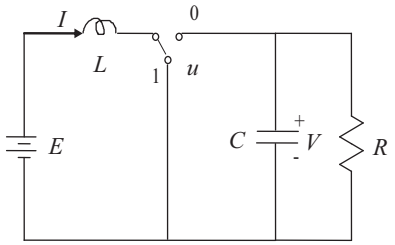


Fig. 4. Boost DC–DC converter.

The main difference of the Boost converter Fig. 1 as compared to the Buck converter Fig. 4 is the location of inductor L.

The dynamic model of the *Boost* converter is given as

$$\dot{x}_1 = -(1-u)\frac{1}{L}x_2 + \frac{E}{L},$$

$$\dot{x}_2 = (1-u)\frac{1}{C}x_1 - \frac{1}{RC}x_2, \tag{2}$$

with $x_1 = I$ and $x_2 = V$.

The topological modification of locating the inductor before the switching element rather than after it as for the Buck converter enables a higher output voltage than the source voltage. However, from a control point of view, the *Boost* converter is more difficult for controlling the output voltage. This lies in the fact that the control u appears in both the current and voltage equations, and both in bilinear fashion. Such a configuration implies a highly nonlinear system with associated difficult control design.

In contrast to the Buck converter the equation for the output voltage of the Boost converter depends on control u directly. Selecting the control in the form $u = (1/2)(1 + \text{sign}(s))$, $s = x_2 - x_2^*$ the sliding mode can be enforced in $s = 0$ with the output voltage equal to the reference input x_2^* [7], if values s and

$$\dot{s} = \frac{1}{2}(1-\text{sign}(s))\frac{1}{C}x_1 - \frac{1}{RC}x_2$$

have opposite signs (x_2^* is assumed to be constant). After the sliding mode starts, the equation for the current x_1 can be obtained via the equivalent control method [8]: $[1-u]_{eq} = (x_2^*/R)(1/x_1)$ is found from equation $\dot{s} = 0$ and substituted into the first equation (2)

$$\dot{x}_1 = -\frac{(x_2^*)^2}{RL}\frac{1}{x_1} + \frac{E}{L} \quad \text{or} \quad \dot{x}_1 = \frac{1}{x_1}\frac{E}{L}\left(x_1 - \frac{(x_2^*)^2}{ER}\right)$$

It is evident that the equilibrium point $x_1^* = (x_2^*)^2/ER > 0$ is unstable, therefore the voltage control method is not acceptable, though it let keep the output voltage equal to the desired value. It means that the zero dynamics with respect to the output voltage are unstable.

Now design the control in compliance with the procedure proposed for the Buck converter. Select discontinuous control such that the current x_1 is equal to x_1^* in sliding mode

$$u = \frac{1}{2}(1-sign(s)) \quad s = x_1 - x_1^*$$

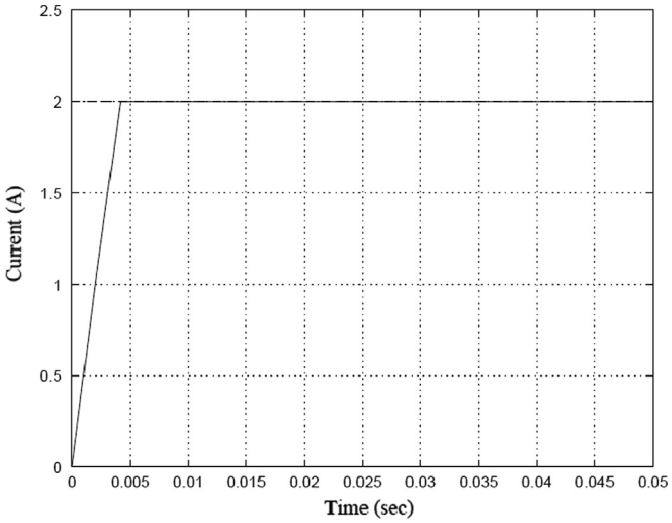


Fig. 5. Current response of a sliding mode controlled buck DC/DC converter.

Again find $[1-u]_{eq} = E/x_2$ from equation $\dot{s} = 0$ and substitute it into the second equation (2)

$$\dot{x}_2 = \frac{Ex_1^*}{C}\frac{1}{x_2} - \frac{1}{RC}x_2 \quad \text{or} \quad \dot{x}_2 = k(x_2)\left(\sqrt{ERx_1^*} - x_2\right), \qquad k(x_2) = \frac{\sqrt{ERx_1^*} + x_2}{RCx_2}$$

The equilibrium point $x_2^* = \sqrt{ERx_1^*}$ is asymptotically stable since $k(x_2) > 0$.

The output voltage tends to the desired value V_d, if $x_1^* = (V_d)^2/ER$.

Note that $u_{eq} = 1-(E/x_2)$, $0 < u_{eq} < 1$ and $x_2 = V_d$ in steady state. Therefore the output voltage is higher than the source voltage. This requirement is essential for a *boost* type DC/DC converter and careful consideration of the initial conditions is required to guarantee the convergence to $s = 0$.

Figs. 5 and 6 show the simulation results of proposed control algorithm for the Boost DC/DC converter. In the simulation, the converter parameters are selected as: $E = 20$ V, $C = 4$ μF, $R = 40\Omega$ and $L = 40$ mH. The desired output voltage is: $V_d = 40$ V. As can be seen from the figures, both the inductor current and the output capacitor voltage converge rapidly to their reference values.

Remark. The voltage tracking problem for a constant reference input can be easily reformulated in terms of the current one. Another particular case with a periodic desired output was studied in Ref. [9]. Generally speaking, it is not a trivial problem for an arbitrary reference input in the system with unstable zero dynamics. The design methodology developed for the systems with a known model of disturbances [10] can be applied directly for nonminimum phase tracking systems. Finally if a reference input is known for some time interval ahead, the efficient design method based on so-called inverse dynamics was offered in Ref. [11]. This approach was used for sliding mode tracking control in systems with unstable zero dynamics [12].

4. Multi-phase DC/DC converters

Finite switching frequency leads to undesirable oscillations with a finite amplitude and frequency. This destructive phenomenon, called “ripple” in power electronics literature, or

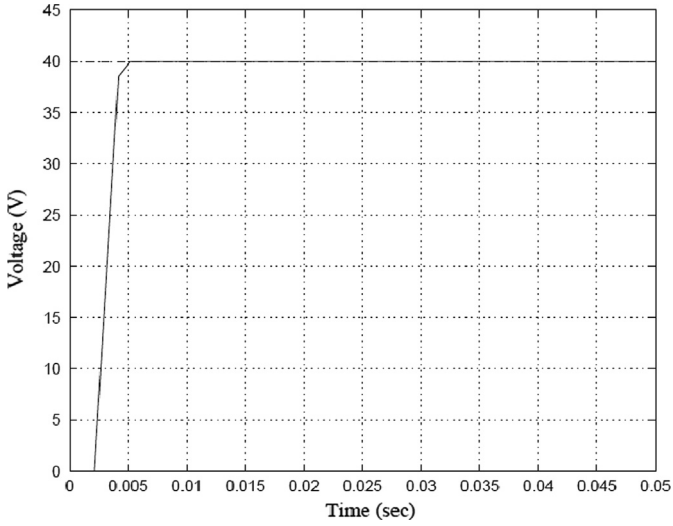


Fig. 6. Voltage response of a sliding mode controlled boost DC/DC converter.

"chattering" in control terminology, may lower control accuracy or incur unwanted wear of mechanical components.

A natural way to reduce chattering is increasing switching frequency. However, it is not always possible due to limitation of switching frequency or losses in power converters. In addition, implementation of sliding modes in power converters results in frequency variations, which is unacceptable in many applications.

4.1. Frequency control

The frequency of chattering and as a result accuracy of a power converter depends on the width of hysteresis loop in switching element and it is selected such that the frequency does not exceed the admissible level for all operation modes. The frequency can be maintained at the desired level equal to the maximum admissible value f_{des} by varying the width of hysteresis loop [13,14].

The design idea can be demonstrated by a first order element implementing pulse width modulation of a continuous signal (Fig. 7).

The function $u_0(x)$ is assumed to be the desired scalar control depending on the state of control system $\dot{x} = f(x.t) + b(x,t)u, \quad (x,f,b \in R^n)$.

As it follows from the equations

$$s = u_0(x) - u, \quad \dot{u} = v = M\text{sign}(s), \quad M > 0,$$
$$\dot{s} = g(x) - M\text{sign}(s), \quad g(x) = [\text{grad}(u_0)]^T(f + bu)$$

sliding mode in the surface $s = 0$ with $u \equiv u_0(x)$ exists if $M > |g(x)|$. If the control is implemented with a hysteresis loop, chattering with oscillation amplitude $A = \Delta/2$ is illustrated in Fig. 8.

Under the assumption that the switching frequency is high enough, the state x can be considered as constant within the time interval $[t_0, t_2]$, and the switching frequency is found as $f = 1/(t_1 + t_2)$, where $t_1 = (\Delta/(M - g(x))), \quad t_2 = \Delta/(M + g(x))$. So, Δ can be selected as

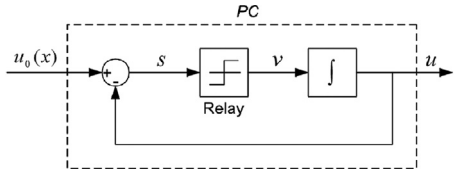


Fig. 7. Sliding mode control for a simple power converter model.

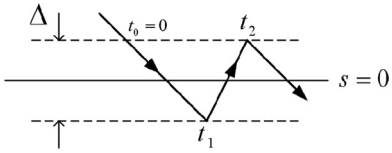


Fig. 8. Oscillation in the vicinity of switching surface.

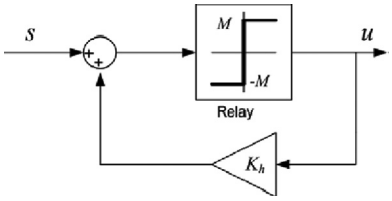


Fig. 9. Implementation of hysteresis loop with width $\Delta = K_h M$.

$\Delta = ((M^2 - g(x)^2)/2M)(1/f_{des})$to maintain the switching frequency at the desired level f_{des}. The hysteresis loop can be implemented by a switching element with gain $K_h$as depicted in Fig. 9.

4.2. Chattering suppression

The challenge is to design a multi-phase converter system based on a given, fixed switching frequency, reducing chattering to a desired level. This may be possible by providing a desired phase shift between phases for any loads and frequencies to implement the "harmonic cancellation" method.

Let us assume now that the desired control is implemented by m power converters, called *multiphase converter,* with $s_i = (u_0/m) - u_i$, $(i = 1, 2, \ldots, m)$ and u_0/m as reference inputs as shown in Fig. 10.

If each power converter operates properly, the output is equal to the desired control $u_0(x)$. Amplitude and frequency in each converter can be found as follows:

$$A = \frac{\Delta}{2}, \quad f = \frac{M^2 - [g(x)/m]^2}{2M\Delta}$$

The amplitude of chattering in u_0 depends on the oscillation in each converter phase, and in the worst case, it can be m times higher than that of each converter. For the system in Fig. 10, phases depend on initial conditions and cannot be controlled.

Suppose that a multiphase converter with m phases is to be designed such that the period of chattering T is the same in each phase, and two subsequent phases have phase shift T/m. Since

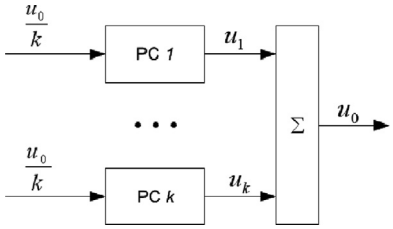


Fig. 10. k-phase converter with evenly distributed reference input.

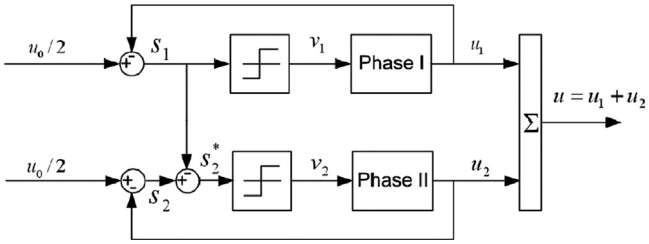


Fig. 11. A controller model with two interconnected phases.

chattering is a periodic time function, it can be represented as Fourier series with frequencies

$$\omega_k = \omega \cdot k, \quad \omega = \frac{2\pi}{T}(k = 1, 2, \ldots, \infty)$$

The effect of k-th harmonic in the output signal is sum of individual outputs from all of phases and can be easily calculated as follows.

$$\sum_{i=0}^{m-1} \sin\left[\omega_k\left(t - \frac{2\pi}{\omega m} i\right)\right] = \sum_{i=0}^{m-1} \mathrm{Im}\left[e^{j(\omega_k t - (2\pi k/m)i)}\right] = \mathrm{Im}(e^{j\omega_k t} Z),$$

$$Z = \sum_{i=0}^{m-1} e^{-j(2\pi k/m)i}$$

To find Z, let us consider the following equation.

$$Z e^{-j(2\pi k/m)} = \sum_{i=0}^{m-1} e^{-j(2\pi k/m)(i+1)} = \sum_{i'=0}^{m} e^{-j(2\pi k/m)i'}$$

$$\text{Since} \quad e^{-j(2\pi k/m)i}\Big|_{i=m} = e^{-j(2\pi k/m)i}\Big|_{i=0}$$

$$Z e^{-j(2\pi k/m)} = Z$$

The function $e^{-j(2\pi k/m)}$ is equal to 1 only If (k/m) is an integer or $k = m.2m, \ldots$, which means that $Z = 0$ for all other cases. This analysis shows that all harmonics except for lm ($l = 1,2,\ldots$) are suppressed in the output signal. As a result, the amplitude of chattering can be reduced to desired level by increasing the number of phases providing a desired phase shift between two subsequent phases

Attempts to apply this idea to Pulse Width Modulation (PWM) have been made such that phase shifts are interconnected and can be controlled, using a transformer with primary and secondary coils in different phases. Alternatively, the phase shift was obtained using delays, filters, or set of triangular inputs with selected delays [15–17].

The approach in this section stems from the nature of the sliding mode control and provides a desired phase shift between phases for any frequency without any additional dynamic elements.

The above analysis lets us outline the design method for suppressing chattering: first, select the width of the hysteresis loop in each phase in compliance with $\Delta = ((M^2 - g(x)^2)/2M)(1/f_{des})$ (or $K_h = \Delta/M$) to have the desired switching frequency. Then select number of phases m and switching functions to provide the desired phase shifts T/m. The last problem will be addressed in the next section.

4.3. Design principle

Let two power converters be implemented as shown in Fig. 11, and the switching function for the second converter is proposed as [18]

$$s_2^* = s_2 - s_1, \quad s_1 = \frac{u_0}{2} - u_1, \quad s_2 = \frac{u_0}{2} - u_2, \quad v_1 = M\,\mathrm{sign}(s_1), \quad v_2 = M\,\mathrm{sign}(s_2^*)$$

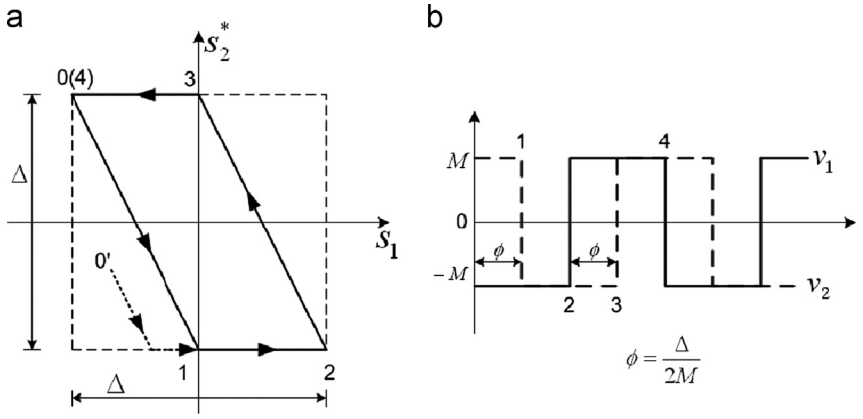


Fig. 12. (a) The system behavior in s-plane, (b) Control of the phase between v_1 and v_2.

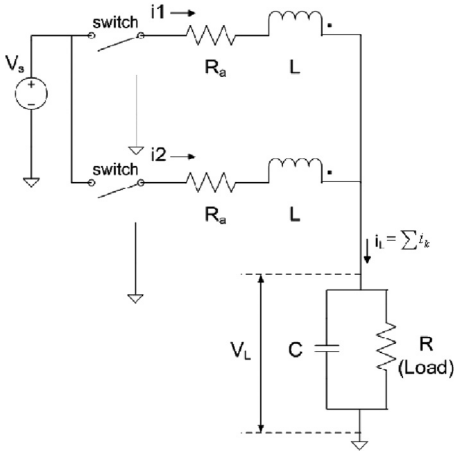


Fig. 13. A two-phase DC to DC converter.

Then, time derivatives of s_1 and s_2^* become

$$\dot{s}_1 = a - M\text{sign}(s_1), \quad a = \frac{g(x)}{2}$$

$$\dot{s}_2^* = M\text{sign}(s_1) - M\text{sign}(s_2^*)$$

The system behavior in the plane s_1 and s_2^* is analyzed with the widths of hysteresis loops for the two sliding surfaces being Δ and $\alpha\Delta$, respectively.

Table 1
Parameter values for simulation.

Parameters	***L* (H)**	***C* (F)**	R_a	R_L	V_s **(V)**
Set *I*	1	1	1	1	12
Set *II*	5×10^{-8}	1×10^{-3}	3×10^{-4}	1×10^{-2}	12

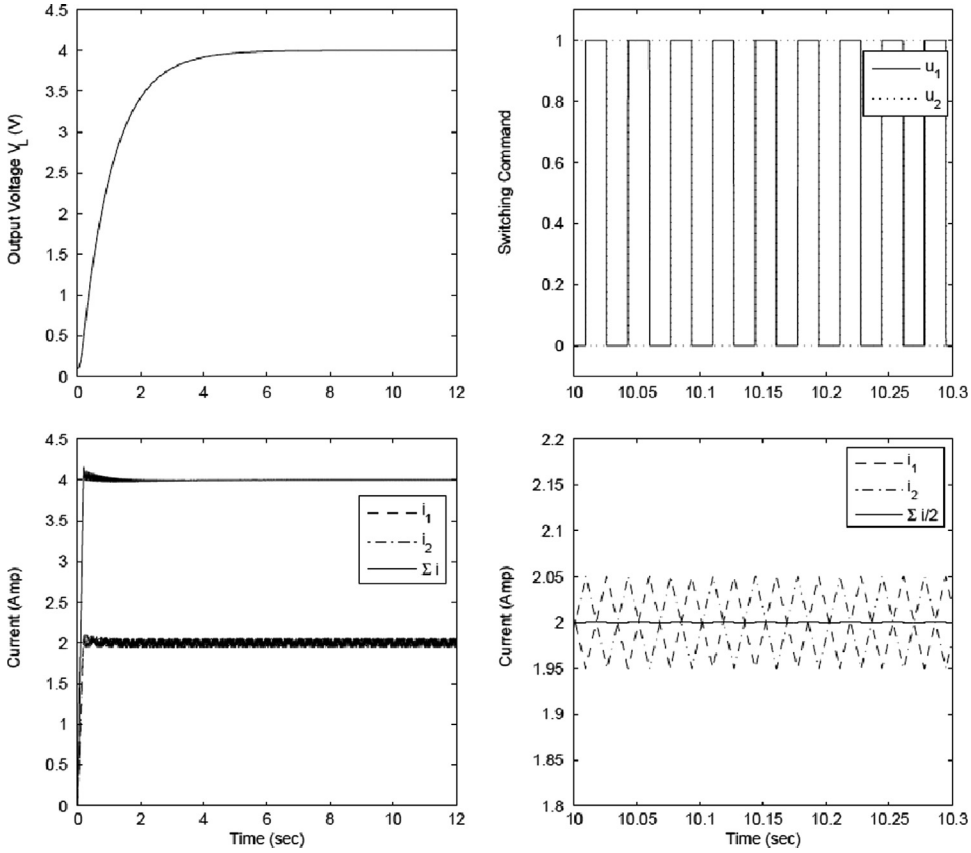


Fig. 14. :Simulation results for the two phase converter with parameters of Set I in MEP_L_tbl1Table 1.

In Fig. 12a,b s-plane is demonstrated for $\propto = 0$ and $\alpha = 1$. For $a \neq 0$ the period of oscillations T can be easily found from the equation with respect to s_1 as follows:

$$T = \frac{\Delta}{M-a} + \frac{\Delta}{M+a} = \frac{2M\Delta}{M^2-a^2}$$

The phase shift for $\propto \neq 1$ becomes

$$\phi = \frac{\propto \Delta}{2M}$$

which is equal to the time for changing s_2^* from $\alpha\Delta/2$ to $-\alpha\Delta/2$ or vice versa.

As follows from this example, the phase shift between oscillations can be selected by proper choice of $\propto$ for any switching frequency without using dynamic elements, e.g., transformers and filters.

The value $\propto$ is calculated to provide the desired phase shift. Since θ must be equal to T/m, $\propto$ can be found as

$$\alpha = \frac{4M^2}{m(M^2-a^2)}$$

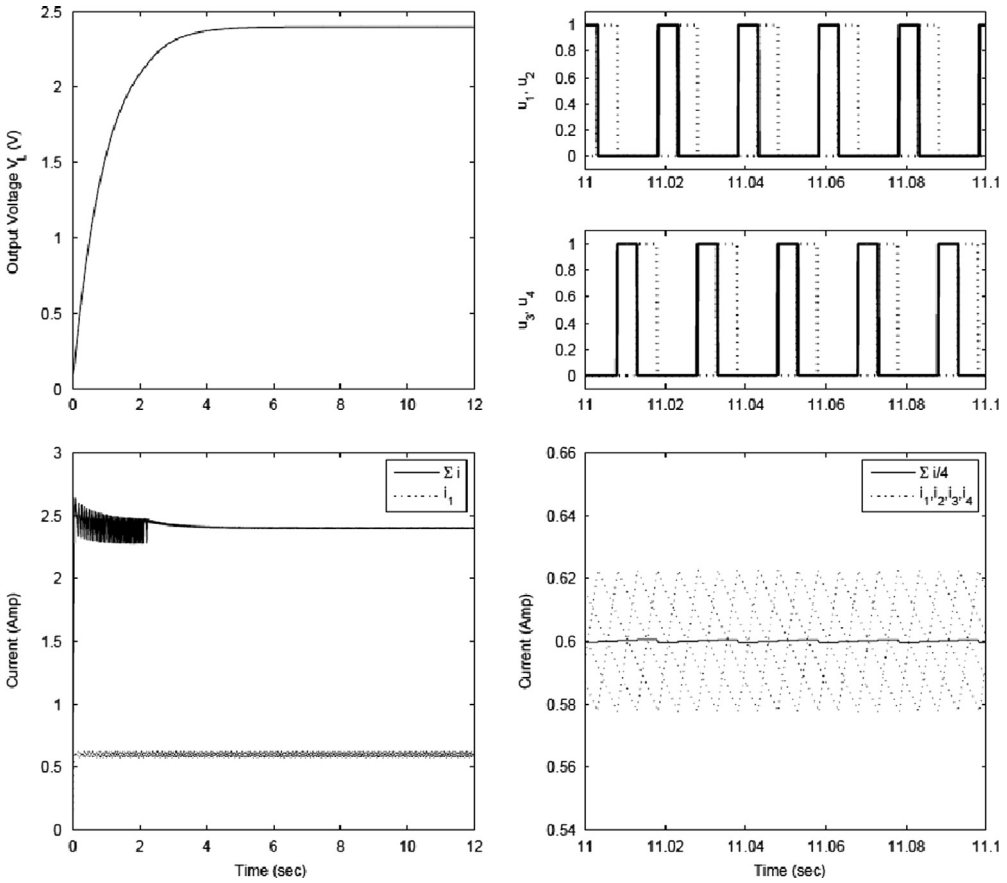


Fig. 15. Simulation results for 4 phases, $V_{ref} = 2.4$ V (parameters of Set I).

The above results may be summarized as the design procedure for the multiphase converter:

- Select the width of hysteresis loop as a state function such that the switching frequency in the first phase is maintained at the desired level.
- Find the parameter $\propto$ as a function of a such that the phase shift between two subsequent phases is equal to $1/m$ of the oscillation period of the first phase

4.4. Simulations and experimental results

The "Harmonic Cancellation" methodology is applied for a DC–DC multiphase buck converter. A converter with 2 phases is depicted in Fig. 13.

The objective is to demonstrate via simulation to what extent chattering can be suppressed in multiphase power converters using the proposed phase shift control methodology. The simulation results are presented for the two parameter sets in Table 1.

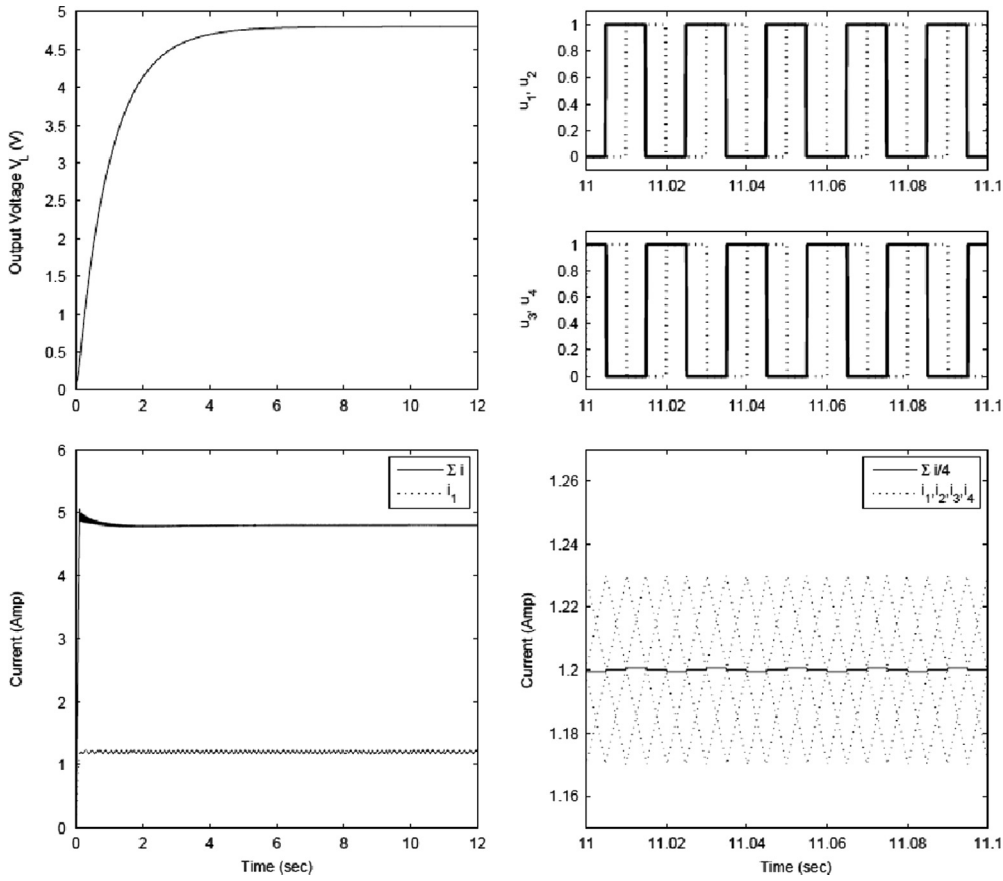


Fig. 16. Simulation results for 4 phases, $V_{ref} = 4.8$ V (parameters of Set I).

For simulation, the governing equations of m-phase converter are assumed as follows:

$$\dot{I}_k = \frac{1}{L}(-I_k R_a + u_k - V_L)(k = 1, 2, \ldots, m), \quad \dot{V}_L = \frac{1}{C}\left(\sum_{k=1}^{m} I_k - \frac{V_L}{R_L}\right) \tag{3}$$

The following control law is used for a two-phase power converter ($m=2$) represented in (3):

$$\dot{s}_1 = I_1 - \frac{I_{ref}}{2}, \quad I_{ref} = \frac{V_{ref}}{R_L}, \quad u_1 = V_s \frac{1-\text{sign}(s_1)}{2}, \quad u_2 = V_s \frac{1-\text{sign}(s_2^*)}{2} \tag{4}$$

where V_{ref} and I_{ref} are the reference voltage input and the corresponding reference load current, respectively. As it is shown in Fig. 14, chattering is suppressed in the output current for the duty cycle equal to 0.5.

The four-phase converter ($m=4$) is simulated with switching frequency control of the first phase by appropriate choice of hysteresis width or hysteresis loop to maintain switching frequency at 50 Hz.

Simulations in Figs. 15–17 are performed for several values of a.

Chattering suppression is observed, and the switching frequency is maintained at the same level.

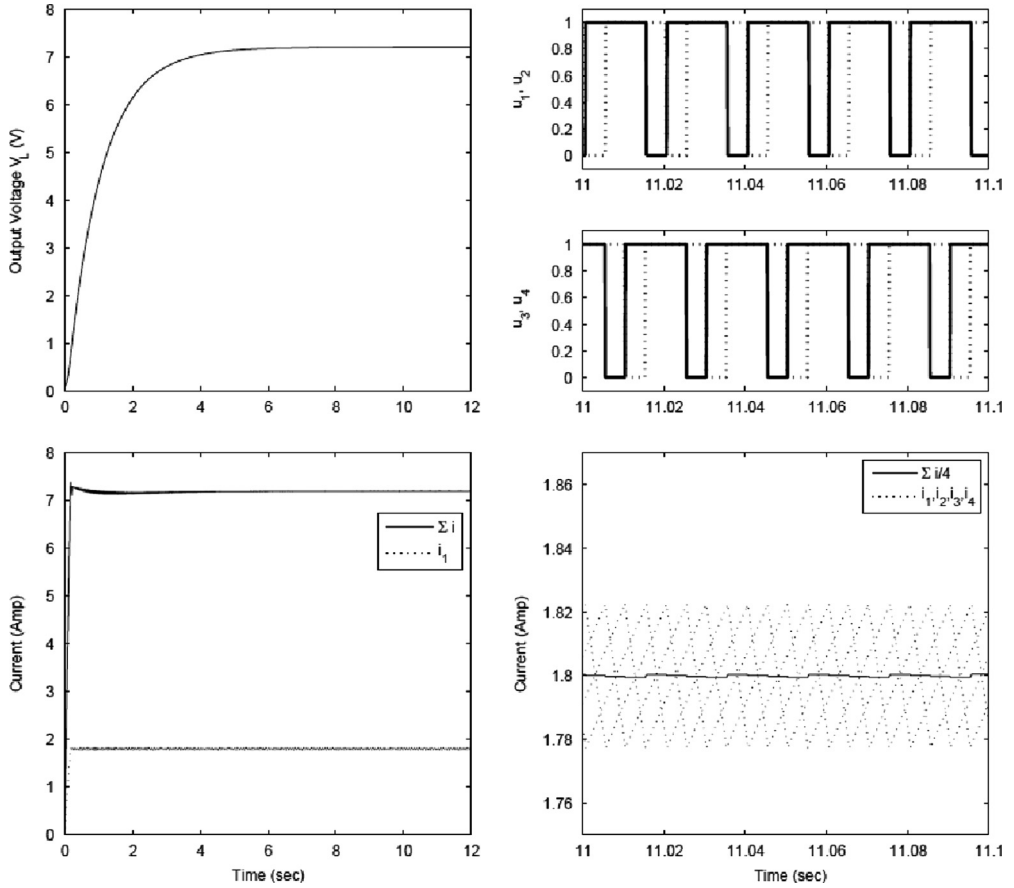


Fig. 17. Simulation result for 4 phases, $V_{ref} = 7.2$ V (parameters of Set I).

For the real-life 4-phase DC to DC power converters with parameters from Set II in Table 1, simulations are performed for different reference inputs. The effect of chattering suppression for reference inputs 3, 6, and 8 V is demonstrated in Figs. 18–20. Note, that the inductance is relatively small in order to have fast converter dynamics. This leads to a high level of chattering in each phase, but it is practically suppressed in the output signal.

For the last case the duty cycle is close to one and relative weight of the fourth harmonic is high enough. According to the harmonic cancellation method this harmonic is not canceled and it can be observed in the fourth diagram.

The results of experiments with four phase converter (Fig. 21) are demonstrated in Ref. [19]. The amplitude of the current chattering is equal to 0.47 A and the oscillation can be observed in the output voltage for a one phase converter (current, output voltage and control command are shown in Fig. 22). As shown in Fig. 23, the chattering amplitude is reduced five times (0.095 A) and chattering is filtered out from the output voltage for the four phase converter. The phase shifts equal to 1/4 of period (Fig. 24) resulted in chattering suppression.

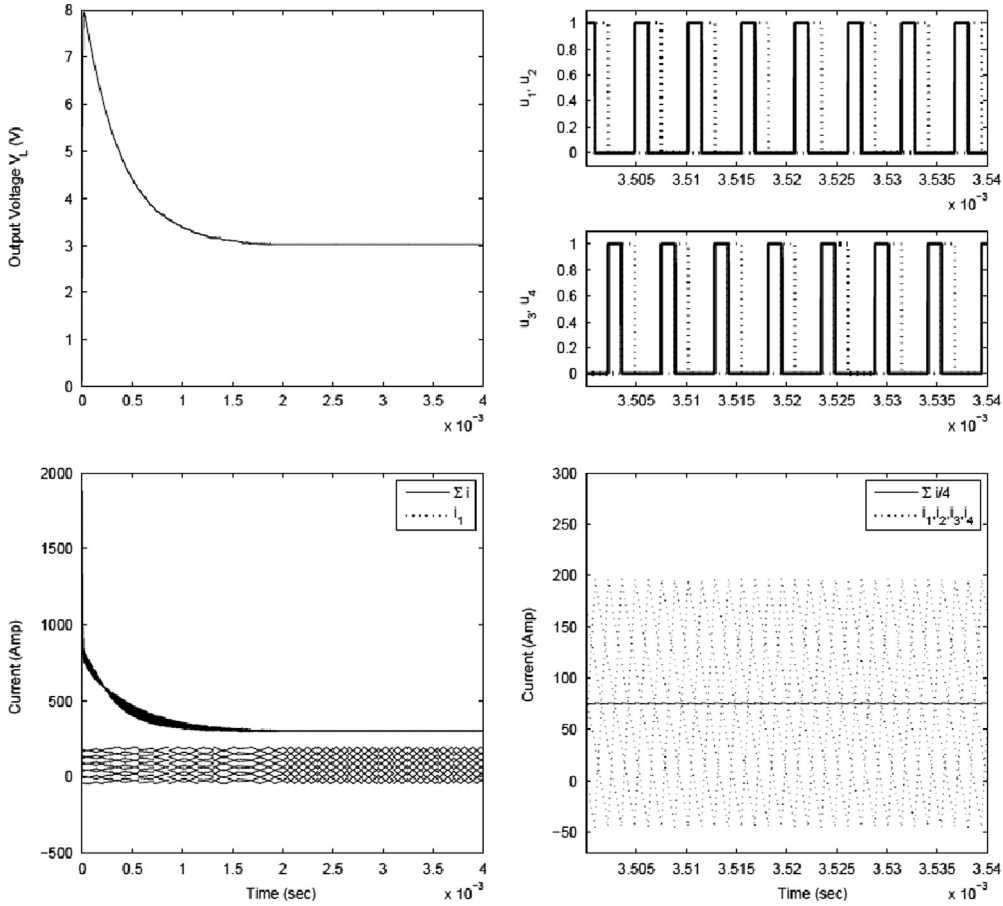


Fig. 18. Simulation result for 4 phases ,$V_{ref} = 3$ V (parameters of Set II).

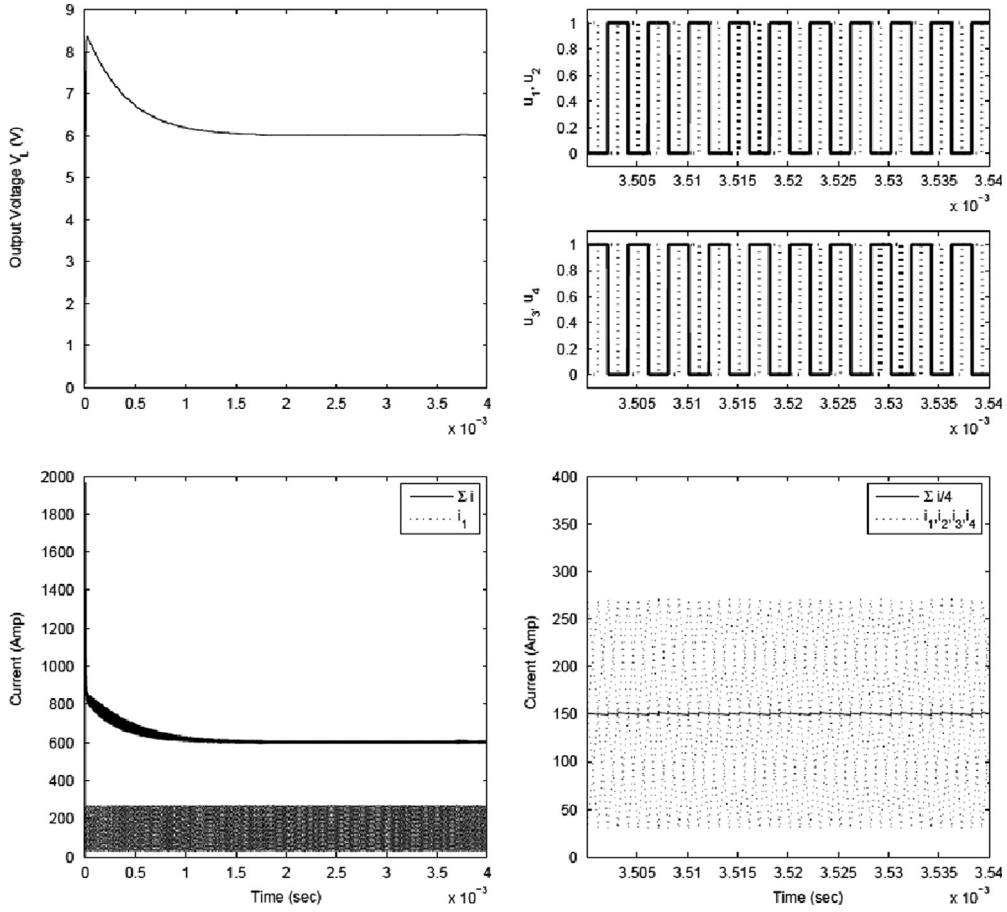


Fig. 19. Simulation result for 4 phases $V_{ref} = 6$ V (parameters of Set II).

5. Converters with a ring configuration

The idea of chattering suppression based on controlling phase shifts can be implemented in multiphase converters with a ring configuration [20] in contrast to the above approach with parallel configuration.

Ring configuration of n boost converters with additional first order filters governed by equations

$$\frac{di_i}{dt} = U_i, \quad U_i = \tfrac{1}{L}\left[V_g - v_c(1-u_i)\right], \quad i = 1, \ldots, n$$

$$\frac{dv_c}{dt} = \tfrac{1}{C}\left[\sum_{i=1}^{n} i_i(1-u_i) - \frac{v_C}{R}\right]$$

$$\tau \dot{z}_i + z_i = i_{i-1}$$

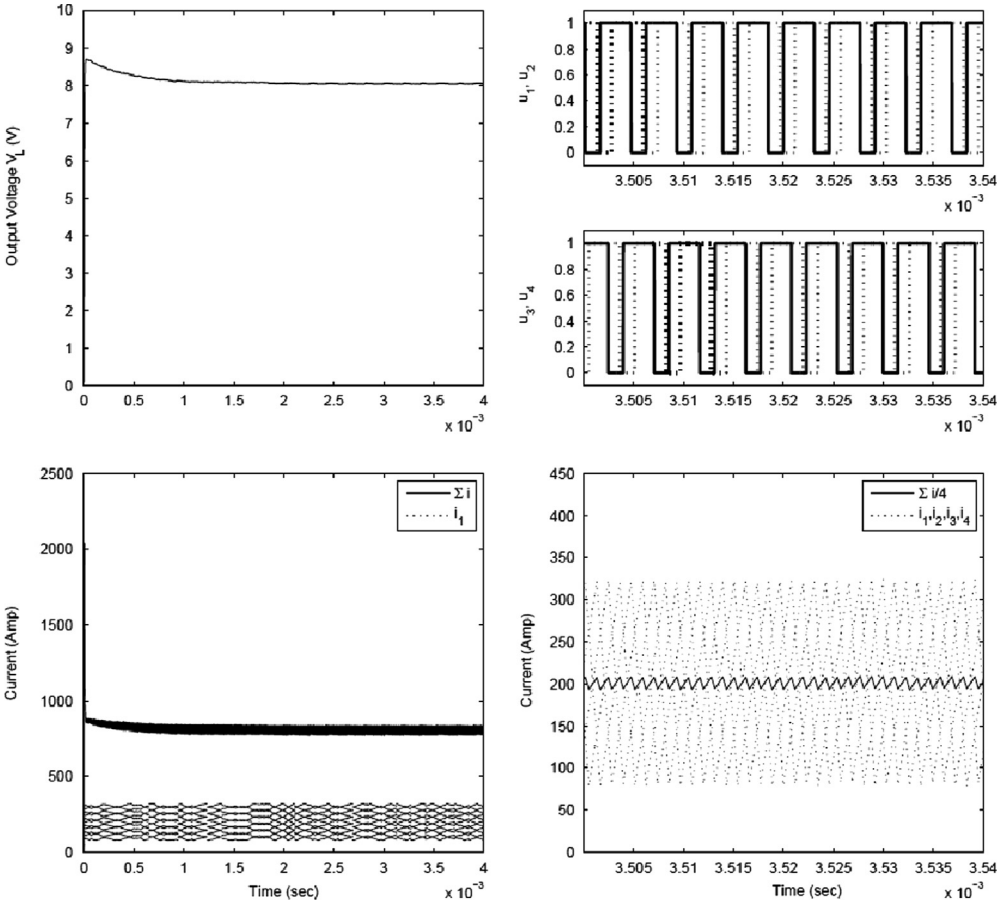


Fig. 20. Simulation result for 4 phases ,$V_{ref} = 8$ V (parameter Set II).

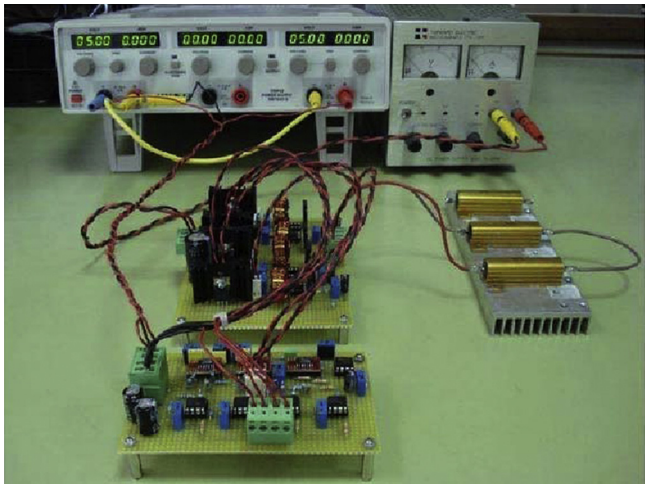


Fig. 21. Four phase experimental setup.

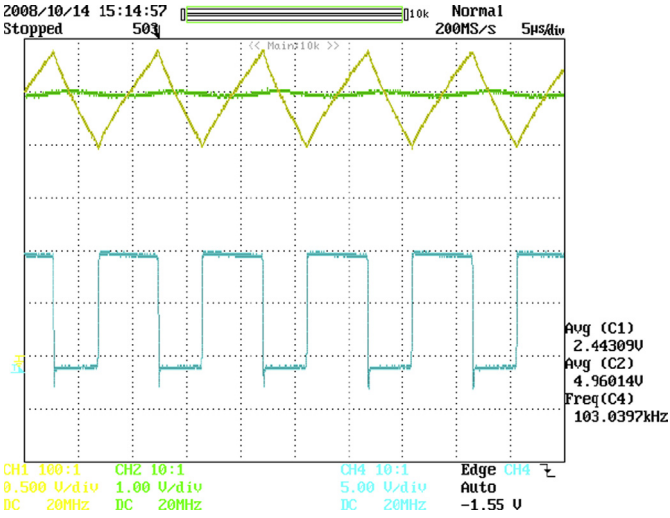


Fig. 22. One phase converter (current, output voltage and control command).

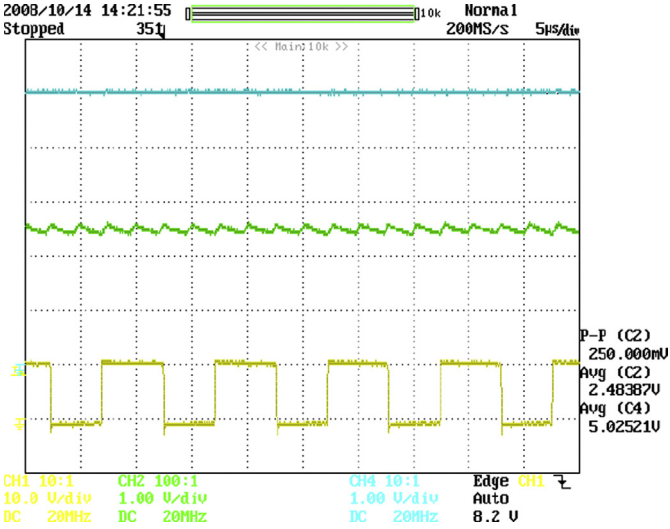


Fig. 23. Four phase converter (current, output voltage and control command).

$$s_i = z_i - bi_i + h, \quad \dot{z}_i = \frac{1}{\tau}(-z_i + i_{i-1}), \quad i_0 = i_n$$

is shown in Fig. 25

Parameter h is selected to have the desired steady state value of the output voltage v_C, parameter b is introduced to stabilize the motion in sliding mode. The input of the first converter i_n is transformed by n similar blocks with the output equal to the input. It means that the periodic component of (chattering) can be shifted by a period (or integer number of periods) and the phase

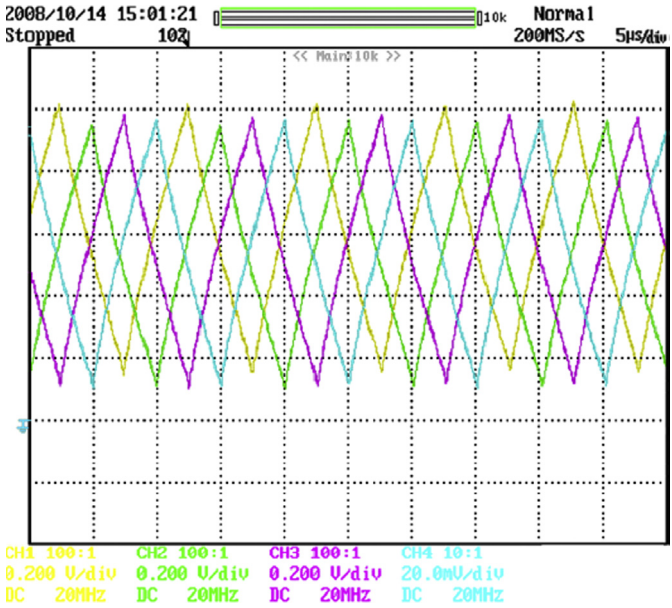


Fig. 24. Phase currents.

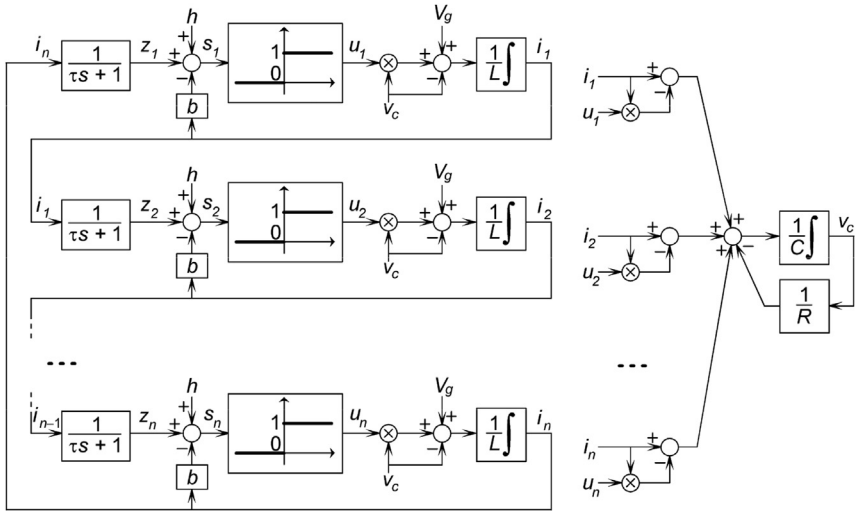


Fig. 25. *n*-phase boost converter with a ring configuration.

shift between two neighbor converters is equal to *1/n* of a period, what is needed for the harmonic cancellation method. As shown in Ref. [20], the periodic oscillation with this phase shifts are asymptotically stable. Fig. 26 shows the currents and the output voltage in 3-phase boost converter with a ring configuration.

According to the harmonic cancellation method the first two harmonics are excluded from the output voltage.

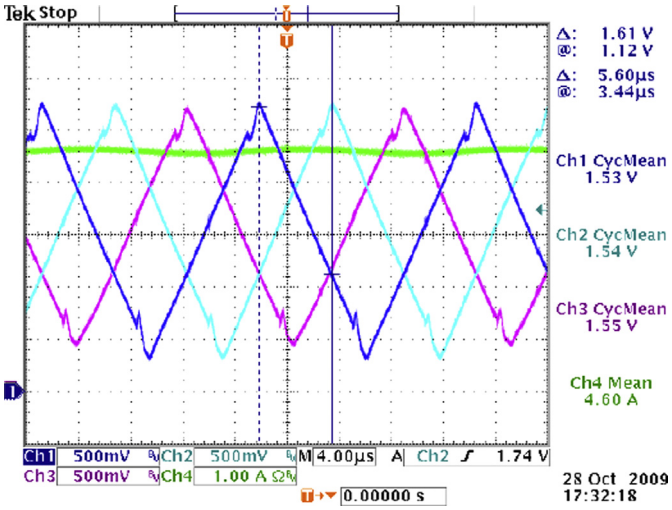


Fig. 26. Phase currents in 3-phase boost converter.

6. Conclusion

Sliding mode control algorithms are presented in the paper along with the equations of sliding motion derived by the equivalent control method. Direct control of current for a buck converter and current and voltage for a bust converter can be implemented. However for current control only stability can be guaranteed. Cascade control for voltage can be implemented by calculating the corresponding reference input for current. Minimal chattering amplitude can be reached by frequency control for converters with maximal admissible switching frequency. Further decreasing frequency to any desired level can be reached in multiphase converters for both parallel and ring configurations based on the harmonic cancellation principle. The solution does not require any additional hardware.

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